

Automorphisms of blow-ups in codimension at least three

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Abstract

Let $\pi: Y \rightarrow X$ be a finite composition of blow-ups of smooth complex projective varieties along smooth centres of codimension at least three. We prove that a finite-index subgroup of $\text{Aut}(Y)$ descends faithfully to $\text{Aut}(X)$. In particular, such blow-ups cannot turn a finite automorphism group into an infinite one. For a blow-up with connected nonempty centre, the numerical pullback space is a hyperplane on which the Hessian determinant of the top intersection form vanishes.

1. Introduction

All varieties are integral and defined over $\mathbf{C}$, and $\text{Aut}(X)$ denotes the group of biregular automorphisms of X . Lesieutre and Litt [2, Theorem 1.6] prove a uniform descent theorem for certain sequences of smooth blow-ups. For a centre of dimension r in an n -fold, their hypotheses are $2r + 3 \leq n$, or $r + 3 \leq n$ together with polyhedrality of the nef cone of the exceptional divisor. They ask whether blow-ups along smooth centres of codimension at least three can turn a finite automorphism group into an infinite one [2, Question 1.8]. We give a negative answer in the smooth complex-projective case.

THEOREM 1.1. *Let*

$$Y = X_m \xrightarrow{\pi_m} X_{m-1} \longrightarrow \cdots \xrightarrow{\pi_1} X_0 = X$$

be a finite sequence of smooth projective varieties, where each π_i is the ordinary blow-up of X_{i-1} along a smooth closed centre, every connected component of which has codimension at least three. Put $\pi = \pi_1 \circ \cdots \circ \pi_m$. There is a finite-index subgroup $G \leq \text{Aut}(Y)$ such that every $g \in G$ admits a unique $\bar{g} \in \text{Aut}(X)$ satisfying

$$(1.1) \quad \pi \circ g = \bar{g} \circ \pi.$$

The homomorphism $G \rightarrow \text{Aut}(X)$, $g \mapsto \bar{g}$, is injective. Consequently, if $\text{Aut}(X)$ is finite, then $\text{Aut}(Y)$ is finite.

We use the top intersection form on numerical divisor classes. Its Hessian determinant is nonzero by the Hodge index theorem. A blow-up along a connected nonempty centre of codimension at least three determines a hyperplane on which this determinant vanishes. There are only finitely many such hyperplanes, so a finite-index subgroup preserves the numerical data of the blow-down. The remaining step is to recover the actual fibres from those data.

2. The intersection discriminant

Let Z be a smooth projective n -fold, with $n \geq 3$, and set $V = N^1(Z)_{\mathbf{R}}$ and $\rho = \rho(Z) = \dim_{\mathbf{R}} V$. Here $N^1(Z)_{\mathbf{R}}$ is the space of real Cartier divisors modulo numerical equivalence. Choose a basis $D_1, \dots, D_\rho$ of V . For $h, u, v \in V$, define (2.1)

$$q_h(u, v) = \int_Z h^{n-2} uv, \quad M_Z(h) = (q_h(D_i, D_j))_{i,j}, \quad \Delta_Z(h) = \det M_Z(h).$$

The integrals denote intersection numbers. If $h = \sum_i x_i D_i$ and $P_Z(x) = \int_Z h^n$, then

$$(2.2) \quad \frac{\partial^2 P_Z}{\partial x_i \partial x_j} = n(n-1)q_h(D_i, D_j), \quad \det \text{Hess } P_Z = (n(n-1))^\rho \Delta_Z.$$

Changing the basis multiplies Δ_Z by a nonzero square, so its zero set is intrinsic.

LEMMA 2.1. *The polynomial Δ_Z is nonzero and homogeneous of degree $d_Z = (n-2)\rho$. Its zero set is preserved by pullback under automorphisms. It contains at most d_Z distinct real linear hyperplanes. Each of these hyperplanes therefore has a stabilizer of index at most d_Z in $\text{Aut}(Z)$.*

Proof. Choose a very ample divisor class a . The degree-two Hodge–Riemann bilinear relations [1, Theorem 7.3.4(b),(c)] say that $\alpha \mapsto \int_Z \alpha^2 a^{n-2}$ is negative definite on the real $(1, 1)$ -classes satisfying $\int_Z \alpha a^{n-1} = 0$. Numerical and real cohomological equivalence agree for divisors: a numerically trivial real divisor has both of these intersection numbers zero, so its Chern class vanishes by this negative definiteness; the converse follows from the cohomological interpretation of intersection. We may thus regard V as the subspace of real $(1, 1)$ -classes spanned by divisor classes. The orthogonal decomposition

$$V = \mathbf{R}a \oplus \left\{ u \in V : \int_Z u a^{n-1} = 0 \right\}$$

has one positive direction, since $a^n > 0$, and a negative definite complement. Hence q_a has signature $(1, \rho-1)$ and $\Delta_Z(a) \neq 0$. The entries in (2.1) are homogeneous of degree $n-2$, which gives the asserted degree.

For $g \in \text{Aut}(Z)$, let T be the matrix of g^* in the chosen basis. Invariance of intersection numbers gives

$$q_{g^*h}(g^*u, g^*v) = q_h(u, v), \quad T^\dagger M_Z(g^*h)T = M_Z(h).$$

Taking determinants yields

$$(2.3) \quad \Delta_Z(g^*h) = (\det T)^{-2} \Delta_Z(h).$$

Finally, a polynomial vanishing on a real hyperplane is divisible by its defining linear form: in coordinates where the hyperplane is $x_\rho = 0$, the constant coefficient in x_ρ vanishes identically. A nonzero polynomial of degree d_Z has at most d_Z linear factors up to nonzero scalars. Pullback permutes the corresponding hyperplanes, proving the index bound. We use the right action $W \cdot g = g^*W$ on hyperplanes. $\square$

3. Blow-ups and descent

PROPOSITION 3.1. *Let $\pi: Y = \text{Bl}_S X \rightarrow X$ be the blow-up of a smooth projective n -fold along a nonempty connected smooth centre of codimension $c \geq 3$. The stabilizer of $\pi^*N^1(X)_{\mathbf{R}}$ in $\text{Aut}(Y)$ has index at most $(n-2)\rho(Y)$ and descends faithfully to $\text{Aut}(X)$.*

Proof. Let E be the exceptional divisor, and write $p: E \rightarrow S$ and $i: E \hookrightarrow Y$ for the natural maps. The morphism p is a $\mathbf{P}^{c-1}$ -bundle, and $\mathcal{O}_E(E)$ restricts to $\mathcal{O}_{\mathbf{P}^{c-1}}(-1)$ on its fibres [3, Tag 0FUT]. We have

$$(3.1) \quad N^1(Y)_{\mathbf{R}} = \pi^*N^1(X)_{\mathbf{R}} \oplus \mathbf{R}[E].$$

Indeed, E is prime. Every other prime divisor F on Y is the strict transform of a Cartier divisor $D = \pi(F)$ on X , so $F = \pi^*D - a_F E$ for some integer a_F . Thus pullbacks and E span $N^1(Y)_{\mathbf{R}}$. Pullback is numerically injective: for an integral curve $B \subset X$, take the reduced closure of a closed point of the generic fibre of $\pi^{-1}(B) \rightarrow B$. This is an integral curve C dominating B with some finite degree $e > 0$, and $(\pi^*D) \cdot C = e(D \cdot B)$. For a line ℓ in a fibre of p , one has $(\pi^*D) \cdot \ell = 0$ and $E \cdot \ell = -1$. This proves the direct sum in (3.1).

Put $W = \pi^*N^1(X)_{\mathbf{R}}$, a hyperplane in $N^1(Y)_{\mathbf{R}}$. For $h = \pi^*a \in W$ and $u \in N^1(Y)_{\mathbf{R}}$, restriction to E gives

$$(3.2) \quad q_h(E, u) = \int_Y (\pi^*a)^{n-2} E u = \int_E p^*(a|_S)^{n-2} i^*u = 0.$$

The last equality holds because $\dim S = n - c < n - 2$. Since $[E] \neq 0$, the form q_h is degenerate for every $h \in W$. Thus $W \subset \{\Delta_Y = 0\}$, and Lemma 2.1 gives

$$(3.3) \quad G_\pi = \{g \in \text{Aut}(Y) : g^*W = W\}, \quad [\text{Aut}(Y) : G_\pi] \leq (n-2)\rho(Y).$$

We next prove regular descent. Choose an ample divisor A on X . If $g \in G_\pi$ and $C \subset Y$ is an integral curve contracted by π , then

$$(3.4) \quad A \cdot \pi_* g_*[C] = (g^* \pi^* A) \cdot C = 0,$$

since $g^* \pi^* A \in W$. If $\pi(g(C))$ were a curve, the left side would be positive. Hence πg contracts every such C . Each nontrivial closed fibre of π is $\mathbf{P}^{c-1}$,

whose closed points are joined by lines. Therefore πg is constant on every closed fibre.

Let Γ be the reduced closed image of $(\pi, \pi g): Y \rightarrow X \times X$. It is integral, and its first projection $q: \Gamma \rightarrow X$ is proper. Every closed fibre of q has just one closed point and is therefore zero-dimensional. Upper semicontinuity of fibre dimension for a proper morphism [3, Tag 0D4I] implies that all its fibres are zero-dimensional: a nonempty closed locus of positive-dimensional fibres would contain a closed point. Thus q is quasi-finite and hence finite [3, Tag 02LS]. Over $X \setminus S$, the image is the graph of $\pi g \pi^{-1}$, so q is birational. Since X is normal, q is an isomorphism [3, Tag 0AB1]. The second projection defines a morphism $\bar{g}: X \rightarrow X$ satisfying (1.1). Applying the same argument to g^{-1} supplies its inverse. Uniqueness and compatibility with composition follow from (1.1) and the surjectivity of π . An element descending to the identity fixes $\pi^{-1}(X \setminus S)$ pointwise and hence is the identity on Y . This proves faithfulness. $\square$

Proof of Theorem 1.1. A smooth centre has finitely many disjoint connected components. Blowing up their union is the same as blowing them up successively: on the open set obtained by removing all but one component, only that component contributes, and these identifications agree on overlaps. The remaining components lie in the isomorphism locus of each preceding step, so their codimensions do not change. Refine the sequence in this way, omit empty centres, and relabel its terms $X_0, \dots, X_m$. If no steps remain, π is an isomorphism and the assertion follows. Otherwise $n = \dim X \geq 3$.

For each step, Proposition 3.1 gives a finite-index subgroup $G_i \leq \text{Aut}(X_i)$ and an injective descent homomorphism $\delta_i: G_i \rightarrow \text{Aut}(X_{i-1})$. Set $H_0 = \text{Aut}(X_0)$ and $H_i = \delta_i^{-1}(H_{i-1})$. The elementary coset inequality gives

$$[G_i : H_i] = [\delta_i(G_i) : \delta_i(G_i) \cap H_{i-1}] \leq [\text{Aut}(X_{i-1}) : H_{i-1}].$$

Induction shows that H_i has finite index in $\text{Aut}(X_i)$ and descends faithfully through the first i steps. Take $G = H_m$. If $\text{Aut}(X)$ is finite, then G is finite, and its finite index makes $\text{Aut}(Y)$ finite. $\square$

Remark 3.2. For the refined nonempty sequence, the proof gives the explicit bound

$$[\text{Aut}(Y) : G] \leq \prod_{i=1}^m (n-2)\rho(X_i) = (n-2)^m \prod_{i=1}^m (\rho(X) + i),$$

where m now counts the connected nonempty centres. The last equality uses (3.1) at each step.

Remark 3.3. The codimension condition enters in (3.2). If $c = 2$ and a is very ample on X , put $\xi = c_1(\mathcal{O}_E(-E))$. Since $p: E \rightarrow S$ has fibre $\mathbf{P}^1$ and

$p_*\xi = 1$, the same calculation instead gives

$$q_{\pi^*a}(E, E) = - \int_E p^*(a|_S)^{n-2} \xi = - \int_S (a|_S)^{n-2} < 0.$$

The mixed pairings vanish because they restrict to $n-1$ divisor factors on S . On the pullback space the form is the nondegenerate form q_a of X . Consequently $\Delta_Y(\pi^*a) \neq 0$ in this case. Uniform descent can indeed fail in codimension two [2, Example 2.5].

References

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